



Mathematics 2 (5EZB0) Lecture 7

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Taylor series

Higher order approximation

Let $f: \mathcal{D} \rightarrow \mathbb{R}$ have continuous $(k + 1)^{\text{th}}$ order partial derivatives. Let $\mathbf{x} \in \mathcal{D}$ and $\mathbf{h} \in \mathbb{R}^d$ be such that the line connecting $\mathbf{x}$ and $\mathbf{x} + \mathbf{h}$ lies in $\mathcal{D}$. Then, there exists $\theta \in [0,1]$,

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \mathbf{h} \cdot \nabla f(\mathbf{x}) + \frac{1}{2} (\mathbf{h} \cdot \nabla)^2 f(\mathbf{x}) + \cdots + \frac{1}{k!} (\mathbf{h} \cdot \nabla)^k f(\mathbf{x}) + \frac{1}{(k+1)!} (\mathbf{h} \cdot \nabla)^{k+1} f(\mathbf{x} + \theta \mathbf{h})$$

Taylor series

Let $f: \mathcal{D} \rightarrow \mathbb{R}$ be a smooth function (all higher order partial derivatives exist and are continuous). If there exists an $r > 0$, such that for all $|\mathbf{h}| < r$, one has $\frac{|(\mathbf{h} \cdot \nabla)^k f(\mathbf{x})|}{k!} \rightarrow 0$, then

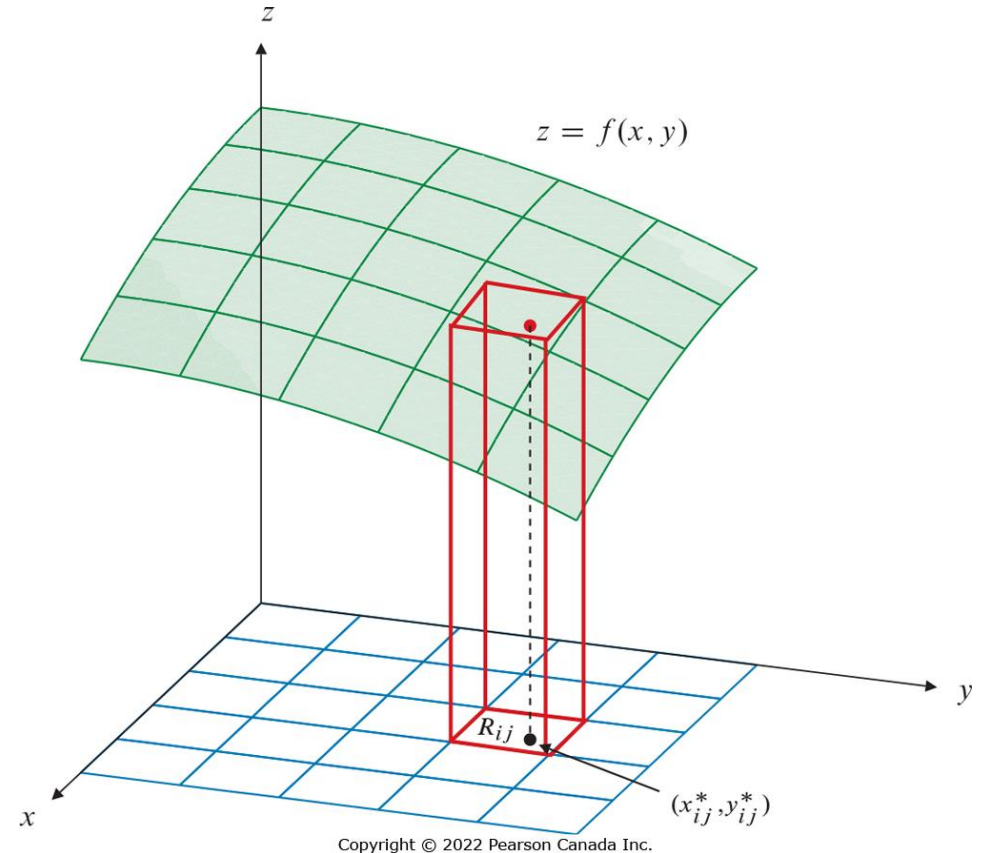
$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \sum_{k=1}^{\infty} \frac{1}{k!} (\mathbf{h} \cdot \nabla)^k f(\mathbf{x})$$

Double integrals

- Consider the rectangles $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ where
$$a = x_0 < x_1 < x_2 < \cdots < x_m = b$$
$$c = y_0 < y_1 < y_2 < \cdots < y_n = d$$
- R_{ij} has area $\Delta A_{ij} := (x_i - x_{i-1})(y_j - y_{j-1})$ and diagonal $d_{ij} > 0$
- consider the partition $\mathcal{P} = \{R_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n\}$ and let $\|\mathcal{P}\| := \max_{1 \leq i \leq m} \max_{1 \leq j \leq n} d_{ij}$
- Now consider the *Reimann sum*: for some $(x_{ij}^*, y_{ij}^*) \in R_{ij} \in \mathcal{P}$,

$$R(f, \mathcal{P}) = \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

- $\int \int_{\mathcal{D}} f(x, y) dA = \lim_{\|\mathcal{P}\| \rightarrow 0} R(f, \mathcal{P})$



Theorem

If $f: \mathcal{D} \rightarrow \mathbb{R}$ is only discontinuous in a set which has *measure 0*, then the function f is integrable in $\mathcal{D}$, and the integral does not depend on the choice of $\mathcal{P}$ and the choice of (x_{ij}^*, y_{ij}^*) .

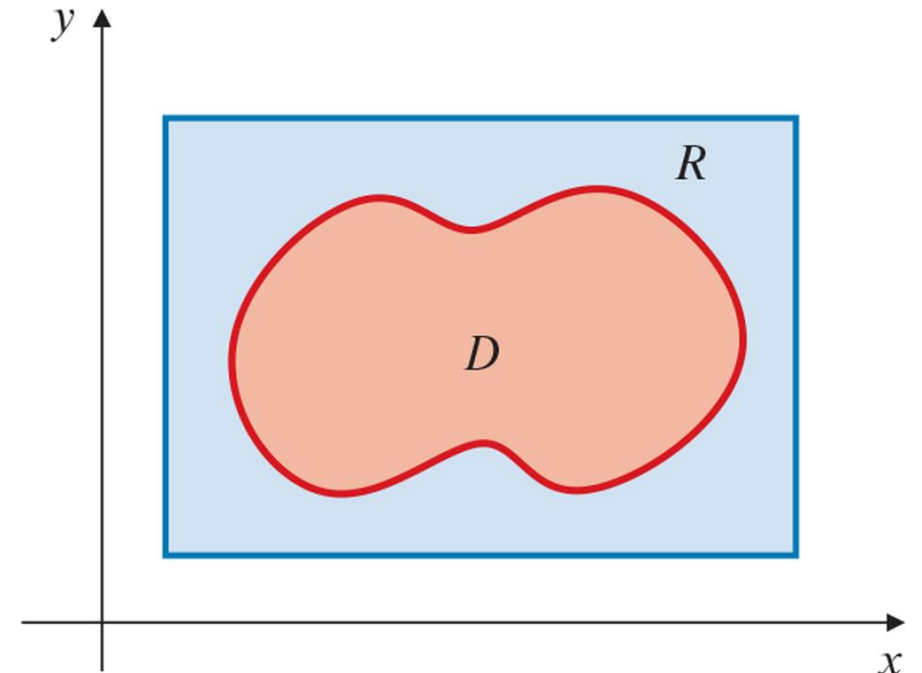
Riemann integrals

- If $\mathcal{D}$ is not a rectangle, then consider a rectangle R surrounding it, and define

$$\tilde{f}(x, y) = f(x, y) \text{ when } (x, y) \in \mathcal{D}, \quad \tilde{f}(x, y) = 0 \text{ otherwise.}$$

Then, if boundary of $\mathcal{D}$ has measure zero, then

$$\int \int_{\mathcal{D}} f(x, y) dA = \int \int_R \tilde{f}(x, y) dA$$



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Properties of integrals

- $\int \int_{\mathcal{D}} f(x, y) dA \geq 0$ if $f(x, y) \geq 0$
- $\int \int_{\mathcal{D}} 1 dA = |\mathcal{D}|$ (Area of $\mathcal{D}$), and $\int \int_{\mathcal{D}} 0 dA = 0$,

Properties of the integrals

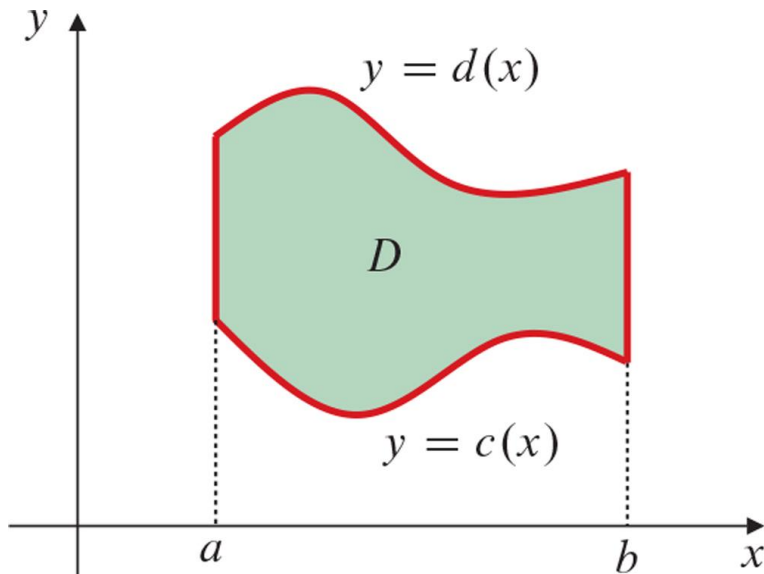
- If $\mathcal{D}$ has 0 measure, then $\int \int_{\mathcal{D}} f(x, y) dA = 0$
- $\int \int_{\mathcal{D}} (\alpha f(x, y) + \beta g(x, y)) dA = \alpha \int \int_{\mathcal{D}} f(x, y) dA + \beta \int \int_{\mathcal{D}} g(x, y) dA$ **(Linearity)**
- $\int \int_{\mathcal{D}} f(x, y) dA \leq \int \int_{\mathcal{D}} g(x, y) dA$ if $f(x, y) \leq g(x, y)$ in $\mathcal{D}$ **(Preserving ordering)**
- $|\int \int_{\mathcal{D}} f(x, y) dA| \leq \int \int_{\mathcal{D}} |f(x, y)| dA$ **(Triangle inequality)**
- Let $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_k$ be disjoint domains. Then **(Additivity)**

$$\int \int_{\cup_{i=1}^k \mathcal{D}_i} f(x, y) dA = \sum_{1 \leq i \leq k} \int \int_{\mathcal{D}_i} f(x, y) dA$$

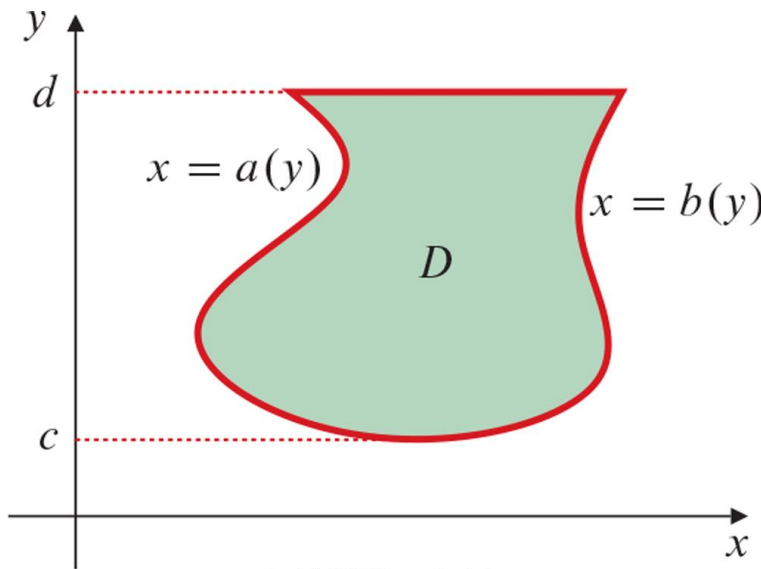
Questions 1

Find the integral $\int \int_{\{1 \leq x^2 + y^2 \leq 2\}} \sqrt{x^2 + y^2} dA$

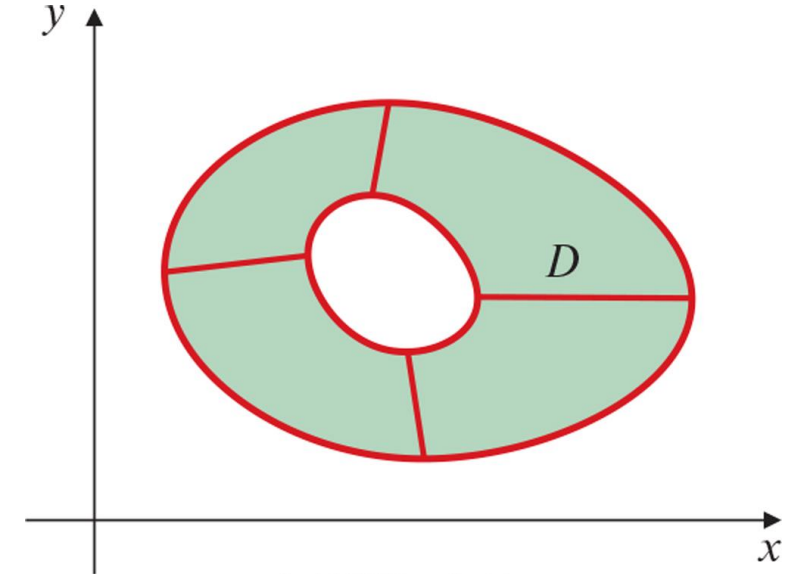
Integrals in Cartesian coordinates: domains



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x -simple domains

Let $c, d: [a, b] \rightarrow \mathbb{R}$ be continuous, and

$$D := \{(x, y): a \leq x \leq b, \\ c(x) \leq y \leq d(x)\}$$

y -simple domains

Let $a, b: [c, d] \rightarrow \mathbb{R}$ be continuous, and

$$D := \{(x, y): c \leq y \leq d, \\ a(y) \leq x \leq b(y)\}$$

Regular domains

Domains that can be decomposed into a finite number of x -simple or y -simple domains.

Examples

Convex polygons, ellipses etc. are both x - and y -simple.

Questions 2

Can you find some non-regular domains in $[0, 1]^2$?

Integrals in Cartesian coordinates: computation

- *x-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_a^b \left(\int_{c(x)}^{d(x)} f(x, y) dy \right) dx$
- *y-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_c^d \left(\int_{a(y)}^{b(y)} f(x, y) dx \right) dy$

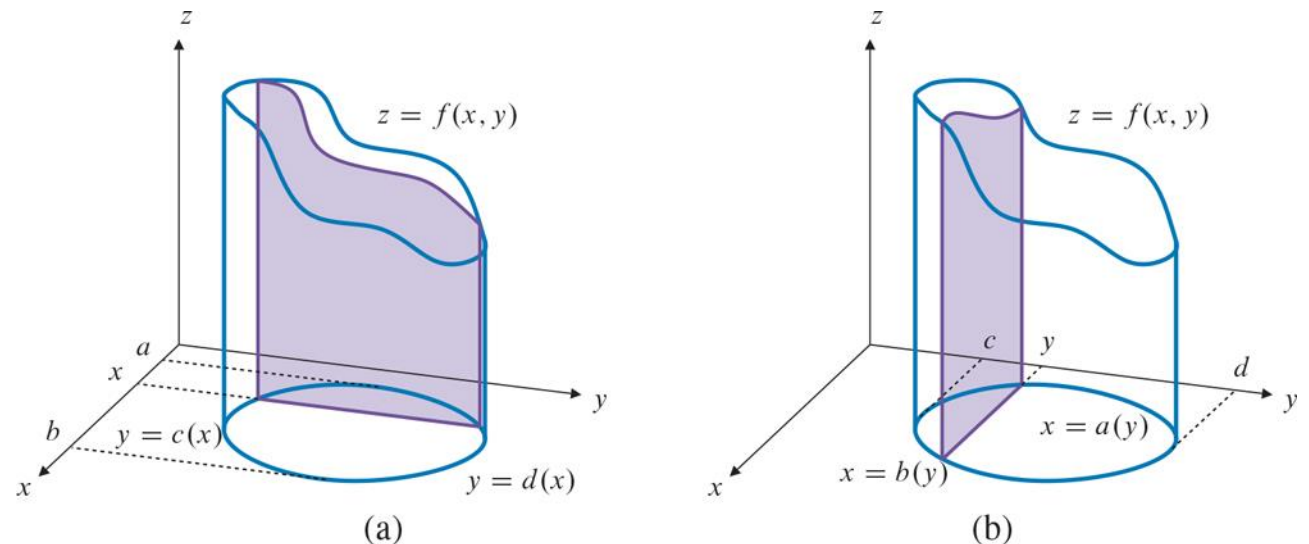
Run **Integral_slice.py** to see how the volume can be calculated using slices

Fubini's Theorem

If $\mathcal{D}$ is both *x-simple* and *y-simple* and $f: \mathcal{D} \rightarrow \mathbb{R}$ is continuous, then both integrals are the same.

Hence, we also write (with the association of dA as $dx dy$)

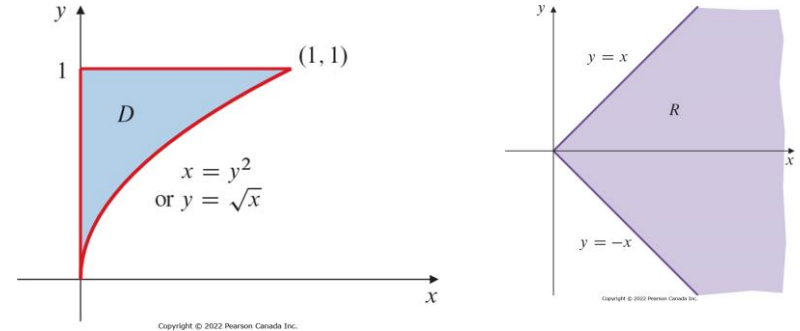
$$\int \int_{\mathcal{D}} f(x, y) dA \quad \text{as} \quad \int \int_{\mathcal{D}} f(x, y) dx dy$$



Integrals in Cartesian coordinates: computation

➤ *x-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_a^b \left(\int_{c(x)}^{d(x)} f(x, y) dy \right) dx$

➤ *y-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_c^d \left(\int_{a(x)}^{b(x)} f(x, y) dx \right) dy$



Fubini's Theorem

If $\mathcal{D}$ is both x -simple and y -simple and $f: \mathcal{D} \rightarrow \mathbb{R}$ is continuous, then both integrals are the same.

Hence, we also write (with also the association of dA as $dx dy$)

$$\int \int_{\mathcal{D}} f(x, y) dA \text{ as } \int \int_{\mathcal{D}} f(x, y) dx dy$$

Questions 3

1. Show $\int_a^b \int_c^d f(x)g(y) dx dy = \left(\int_a^b f(x) dx \right) \left(\int_c^d g(y) dy \right)$

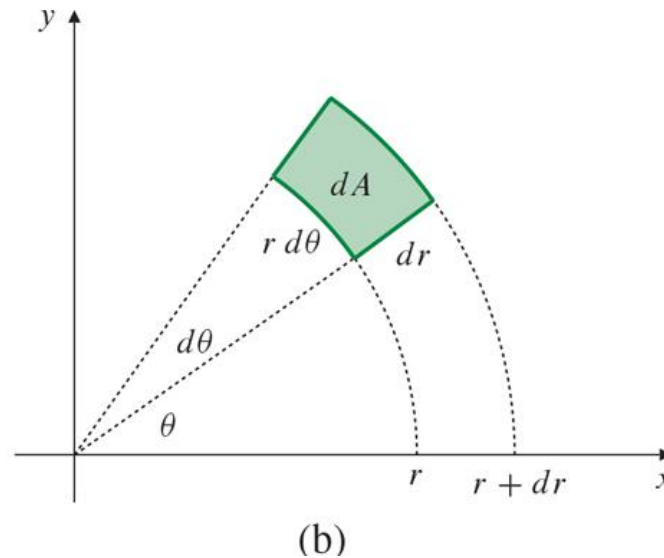
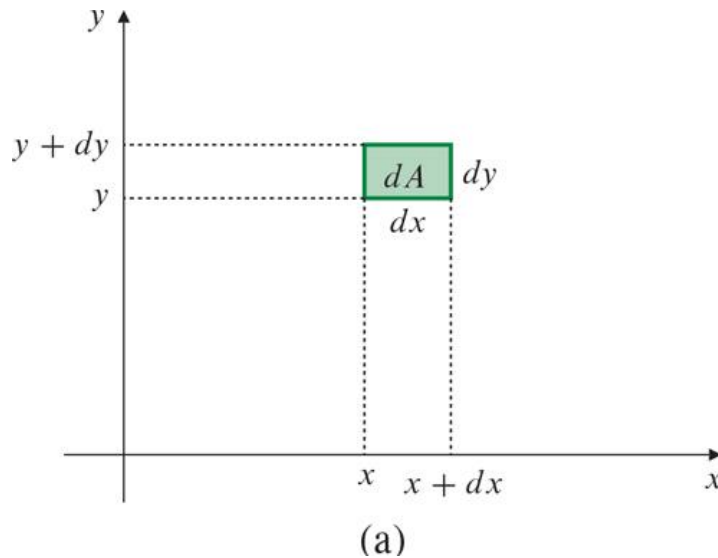
2. Evaluate the integral $\int_0^1 \int_{\sqrt{x}}^1 e^{y^3} dx dy$

3. Evaluate the integral $\int \int_{\{x>|y|>0\}} e^{-x^2} dx dy$

3. Find the volume of the sphere using: $\int \sqrt{1-x^2} dx$
 $\int \int_{\{x^2+y^2 \leq 1\}} \sqrt{1-x^2-y^2} dx dy = \frac{1}{2} \sin^{-1} x + \frac{1}{2} x \sqrt{1-x^2} + c$

Integrals in polar coordinates: domains

For functions that are easier to write in terms of cylindrical coordinates, can be integrated differently.



Let

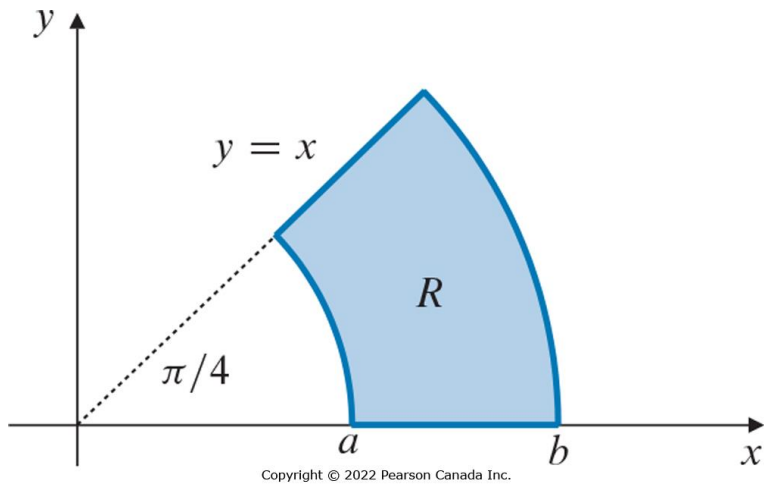
$$f(x, y) = f(r \cos \theta, r \sin \theta) = h(r, \theta)$$

Noticing that an infinitesimal dA can be represented by $dx dy$, but also by $r dr d\theta$, one has

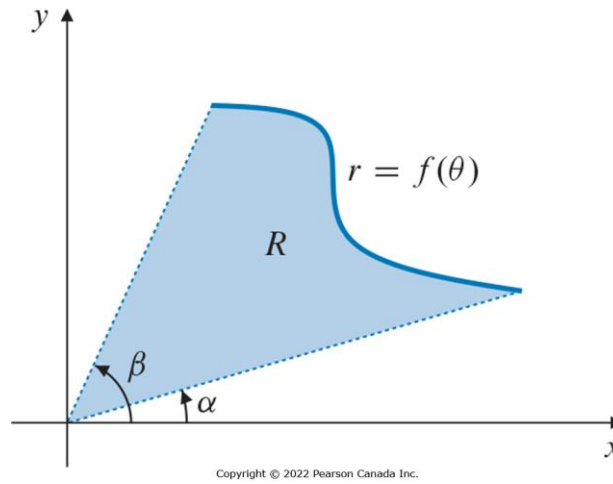
$$\begin{aligned} \iint_{\mathcal{D}} f(x, y) dA &= \iint_{\mathcal{D}} f(x, y) dx dy \\ &= \iint_{\mathcal{D}} h(r, \theta) r dr d\theta \end{aligned}$$

Integrals in polar coordinates: computing

For functions that are easier to write in terms of cylindrical coordinates, can be integrated differently.



$$\begin{aligned} & \int \int_R h(r, \theta) r dr d\theta \\ &= \int_a^b \left(\int_0^{\pi/4} h(r, \theta) d\theta \right) r dr \end{aligned}$$



$$\begin{aligned} & \int \int_R h(r, \theta) r dr d\theta \\ &= \int_\alpha^\beta \left(\int_0^{f(\theta)} h(r, \theta) r dr \right) d\theta \end{aligned}$$

Let

$$f(x, y) = f(r \cos \theta, r \sin \theta) = h(r, \theta)$$

Noticing that an infinitesimal dA can be represented by $dx dy$, but also by $r dr d\theta$, one has

$$\begin{aligned} \int \int_{\mathcal{D}} f(x, y) dA &= \int \int_{\mathcal{D}} f(x, y) dx dy \\ &= \int \int_{\mathcal{D}} h(r, \theta) r dr d\theta \end{aligned}$$

Questions 4.1 Find the volume of a cone of height h and radius R ? What is a sphere's volume?

Questions 4.2 Prove that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$